TASK 9 – JUNIT TESTING

PROBLEM STATEMENT:

Develop a **Java Calculator application** and perform **unit testing using JUnit 5**.

The Calculator class should provide the following operations:

1. **Addition** – Add two integer numbers.
2. **Division** – Divide one integer by another.
3. **Division by Zero Handling** – Throw an IllegalArgumentException when division by zero is attempted.

Create a **JUnit 5 test class named CalculatorTest** to verify these operations using:

- assertEquals() to test addition and division.
- assertThrows() to test division by zero.

Run the test cases in Eclipse using **Run As → JUnit Test** and verify that all test cases pass successfully. Also demonstrate a **failed test case** by providing an incorrect expected output and observe the failure in the JUnit result window.

SOURCE CODE:

Calculator code:

```
package junitDemo;


public class Calculator {


   public int add(int a, int b) {
      return a + b;
   }
```

```
    public int divide(int a, int b) {
        if (b == 0) {
            throw new IllegalArgumentException("Division by zero");
        }

        return a / b;
    }
}
```

CalculatorTest code:

```
package junitDemo;

import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.api.Test;

public class CalculatorTest {

    Calculator cal = new Calculator();

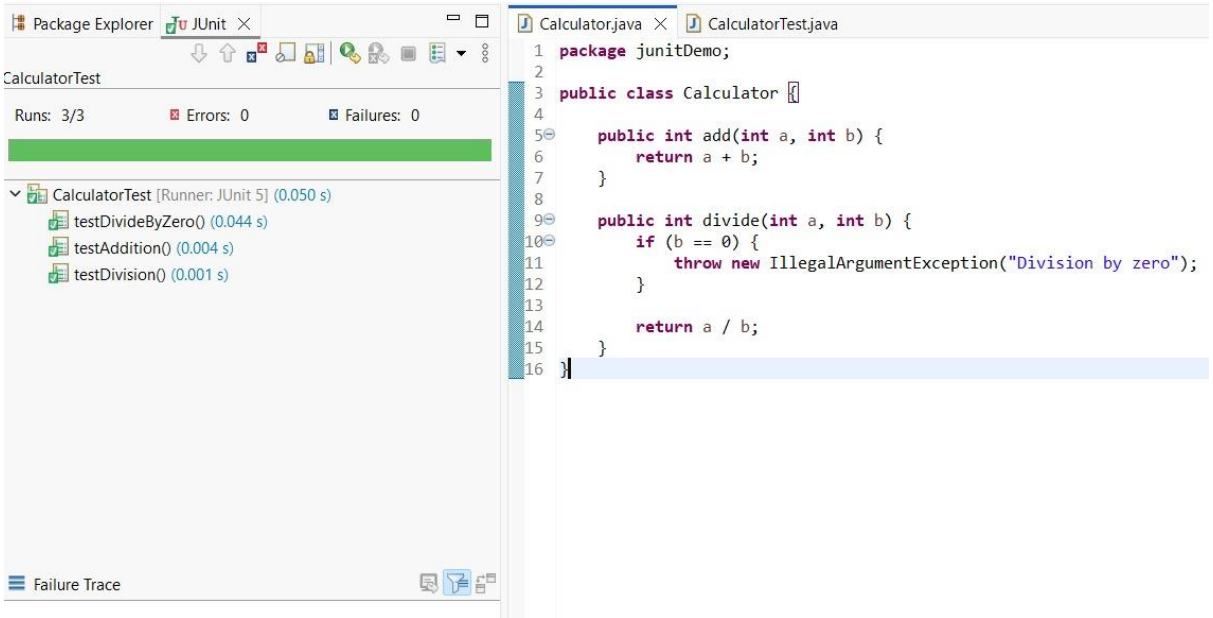
    @Test
    void testAddition() {
        assertEquals(30, cal.add(10, 20));
```

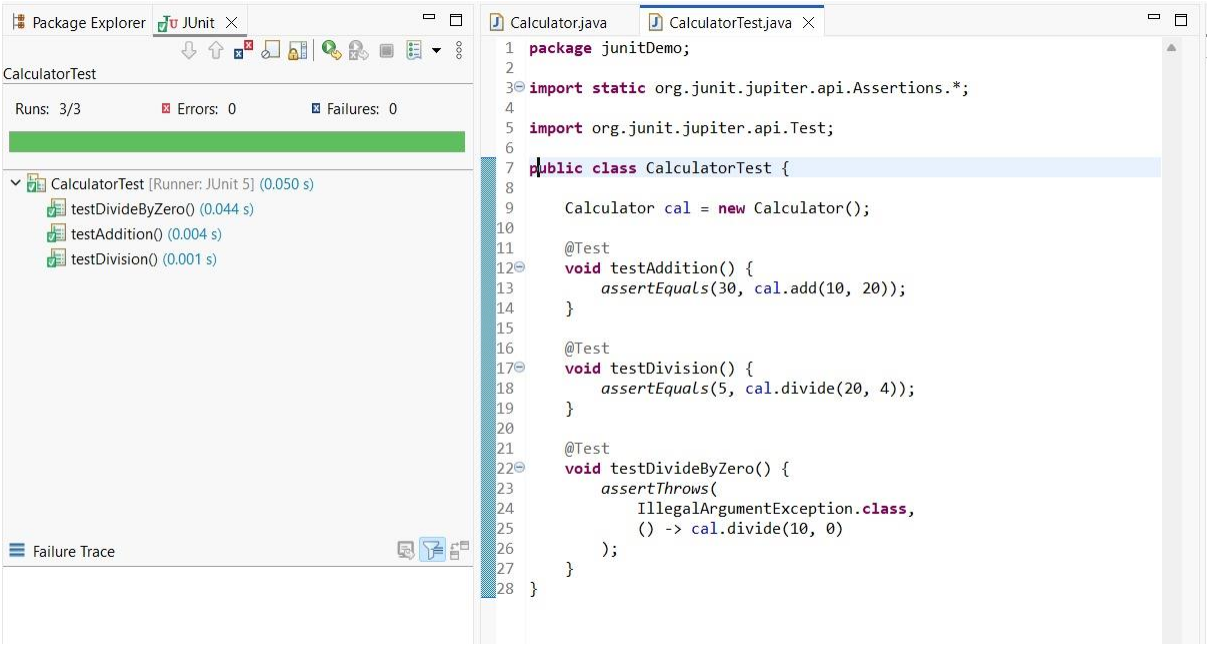
```
    }

    @Test
    void testDivision() {
        assertEquals(5, cal.divide(20, 4));
    }

    @Test
    void testDivideByZero() {
        assertThrows(
            IllegalArgumentException.class,
            () -> cal.divide(10, 0)
        );
    }
}
```

OUTPUT:




Package Explorer
JUnit
CalculatorTest
Runs: 3/3
Errors: 0
Failures: 0
CalculatorTest [Runner: JUnit 5] (0.050 s)
testDivideByZero() (0.044 s)
testAddition() (0.004 s)
testDivision() (0.001 s)
Failure Trace
Calculator.java
CalculatorTest.java
1 package junitDemo;
2
3 import static org.junit.jupiter.api.Assertions.*;
4
5 import org.junit.jupiter.api.Test;
6
7 public class CalculatorTest {
8
9 Calculator cal = new Calculator();
10
11 @Test
12 void testAddition() {
13 assertEquals(30, cal.add(10, 20));
14 }
15
16 @Test
17 void testDivision() {
18 assertEquals(5, cal.divide(20, 4));
19 }
20
21 @Test
22 void testDivideByZero() {
23 assertThrows(
24 IllegalArgumentException.class,
25 () -> cal.divide(10, 0)
26);
27 }
28 }